

Developing Distance Learning Monitoring Dashboard with Google Sheet: An Approach for Flexible and Low-Price Solution in Pandemic Era

Amalia Rahmah
Department of Information System
STT Terpadu Nurul Fikri
Depok, Indonesia
amaliarahmah2@gmail.com

Pristi Sukmasetya
Department of Informatics Engineering,
Universitas Muhammadiyah Magelang
Magelang, Indonesia
pristi.sukmasetya@ummgl.ac.id

Muh Syaiful Romadhon
Department of Information System
STT Terpadu Nurul Fikri
Depok, Indonesia
syaiful@nurulfikri.ac.id

Ahmad Rio Adriansyah
Department of Informatics
STT Terpadu Nurul Fikri
Depok Indonesia
ahmad.rio.adriansyah@gmail.com

Abstract—The Covid-19 pandemic has prompted many colleges and students to conduct distance learning (PJJ), unprepared. Thus, it is necessary to carry out continuous evaluation and improvement in a faster period, monthly or weekly. Therefore, an approach is needed to help the campus stay on track, and still maintain the quality of learning. It is performance monitoring. However, building a monitoring dashboard requires many resources. This study proposes a design solution using Google Sheet, which is flexible, easy to custom, and does not require extensive resources in terms of human resources and costs. The dashboard must provide information that can guide the campus to optimize the distance learning process. Therefore, this study adopts Objective Key Results (OKR), a performance measurement concept using a clear and transparent set of goals, future-oriented, but still realistic. The research consisted of five steps: determining the research scope, OKR analysis, determining the type and mechanism of data input, designing a dashboard, and evaluation. The analyzed OKR consists of 4 objectives and 27 key results (KR). Those KRs are then shown in a monitoring dashboard design based on Google sheet, consisting weekly, semester, and student dashboard. The results of this design will then be used as a monitoring guide for the implementation of distance learning at STT Terpadu Nurul Fikri in the following semester.

Keywords— *google sheet, google form, distance learning, performance monitoring, monitoring dashboard, OKR, objective key results, education in pandemic*

I. INTRODUCTION

As we know, the pandemic has prompted the implementation of distance learning. Bao said it is a disruptive shift with moving all learning activities to become online [1]. The campus and students, including the new ones, must participate, regardless of whether they are ready. It is even more challenging for students experiencing a transition from senior high school to higher education. Likewise, campuses do not yet understand the characteristics or limitations faced by new students who have just entered college. Therefore, there must be many adjustments and treatment interventions in the lecture process to optimize learning in its implementation.

The previous semester observations and student and lecturer representatives' input determine strategies related to this issue. Though, there is an absence of data covering all learning activities as a basis for decision making. On the other hand, data gathering, and processing require much effort and resources, both in energy, cost, and time. Moreover, during a pandemic, resources are more limited than before.

Therefore, we need a flexible monitoring solution with minimal resources to support the optimal implementation of PJJ on campus. A tool widely used in the campus environment is familiar in its use, namely Google Sheet and Google Form. Google Sheet is powerful enough to process data at a certain capacity and flexible enough to visualize data. Based on periodic evaluation activities at the STT Terpadu Nurul Fikri, Google Sheet can accommodate three years of semester of periodical evaluation, within less than 1000 students. This encourages the reuse of Google Sheets and Google Forms as a medium for real-time data visualization and periodically updated data that are flexible and customizable, do not take a long time (time), and cost.

Thus, this study aims to design a monitoring dashboard that covers the design of data input, processing, and visualization. This objective is summarized in two research questions. The first one is how Google Sheet can provide information to support decision making in optimizing the implementation of PJJ in the pandemic era. Given the uncertainty conditions, it is necessary to determine priorities for the things that need to be optimized. Therefore, this study will adopt the OKR concept that supports the adjustment of the focus of activity implementation, not only on processes and outputs but also on expected outcomes. OKR stands for objectives and key results, describing your mission-supporting goal and sets a deadline for achieving it [2]. This adoption of this approach aims to adjust the implementation of PJJ, and as guidance to maintain the learning quality, without adding more works for students or lecturers. It conveys the second research question, namely, how to use the OKR concept in the education field, especially on higher education PJJ implementation in the pandemic era.

II. RATIONALE

A. Google Sheet for Dashboard

Google Sheets is an online-based application developed by Google [3]. The application serves to create and format spreadsheet documents in real time via the internet network [4]. Creating dashboards as well as data storage is easier with the Google Sheets application [5]. Many functions and formulas make it easier to process data; even in Google Sheets, programming scripts can also be inserted if needed [6]. Google Sheets is more efficient in terms of use because it is cloud-based, compared to traditional applications installed on the desktop even though it is more familiar [7].

Google Sheets has several advantages for saving automatically and having editing history without having to connect with other applications [8]. Some of its excellent features are able to present pivot tables and diagrams as well as functions automatically and manually. However, when there are many data rows, it requires a faster connection and cannot be done offline with limited columns to a maximum of 18,278 columns and 200 sheets in one file and limited by 50,000 characters [9].

This cloud-based technology is also the main driver of using Google Sheet for monitoring dashboard development. This technology allows simultaneous access to all related parties to the same dashboard page. Also, PJJ and work from home (WFH) create the need for a more synchronized file management system and real-time information. It eliminates the need for data coordination and consolidation, as it is usually done in a typical work from office setting.

In a literature, the dashboard is built as a business intelligence using a dimensional data model taken from a learning management system. The used tool is QlikView. This solution is suitable for large-scale data to find new insights to support the strategy [10]. However, the requirement in this research is for real-time monitoring with targeted insights. In other literature, there is also a dashboard framework for higher education (HE), which describes what components should be included in the HE's monitoring dashboard [11]. This study only focuses on the learning and teaching assessment section.

A performance dashboard is used for a decision support system, where monitoring is crucial [12]. Monitoring involves metrics and KPI as objects to be observed to find problem finding and solver. The dashboard consists of fact-finding modules, problem-solving modules, data management modules, visual user interface modules, and stakeholders. Google Sheet has been used as a dashboard in other literature, but it focuses on handling campus facilities with asynchronized fact-finding [13].

The convenience and low cost are why Google Sheet has become a proposed solution in this study. STT Nurul Fikri, where the research is conducted, has the characteristic which manages less than 1000 students. Therefore, every solution, including a dashboard, can be built with simple available tools. Even if there is a need for more storage later, the required costs are still affordable, compared to the costs incurred for the construction of monitoring dashboards using other tools or development on the existing academic information system (AIS) framework.

B. OKR for Agile Distance Learning Monitoring

Before the pandemic, the idea of smart education and a smart learning environment had already been echoed. Smart education is an education that can encourage students to improve the quality of their learning [14]. One of them is using existing data to recommend learning experiences that suit the conditions and needs of students. In global education, the new paradigm of smart education facilitates and helps students solve problems in the long term [14]. Learning must be supported by supportive technology and cannot be separated from the teaching process [15] [16]. On the other side, a smart environment is encouraged to realize smart education in increasing the efficiency and effectiveness of learning [17]. Thus, to promote the learning process's continuity, it is necessary to support improvement recommendations regarding facilities, teaching and learning process, and understanding as a learning outcome.

The implementation of smart education and learning environment is closely related to the use of AI to provide various recommendations for improving the learning experience. Though to realize this idea, there is a need to start by gathering the data and initiate data processing and visualization, with manageable volume. Also, there is a need to build consistency in the data gathering process. This emerging idea is based on the consideration that there was no initiation towards smart education previously. It underlies the notion of a monitoring dashboard, equipped with integrated input media and data processing. The conditions that require a quick response with all resource limitations are in line with Google Sheet and Google Forms technology solutions.

The pandemic has forced the implementation of PJJ with various compromises to keep it running, though it is not ideal. In the previous semester, many decisions needed to be made to adjust the situation. However, various priorities and directions need to be considered in decision making. This complexity can be looked at from three viewpoints, students, lecturers, and the learning environment. Each viewpoint carries an issue to be considered. There are students' digital literacy and infrastructure support, lecturers' capability to deliver online learning, and related policies.

These various concerns conflict with each other to determine the strategy. It is necessary to find an optimal point, namely, the priority focus. It can be realized by setting priorities based on the goals to be achieved in online lectures. This goal setting is necessary because, generally, the goals of study programs and campuses are still generic and in pre-condition non-pandemic situations. There needs to be an adjustment/reduction in goals specifically related to the administration of PJJ in the pandemic era. Therefore, there is a need for a thinking framework about how strategies and decisions should be made, with clear reasoning. It encourages the use of OKR in this study, which provides a framework for determining an organization's direction to keep running and growing based on the set of goals.

OKR has been widely adopted by various high growth performance companies such as Intel, Google, and Amazon. In OKR, goals cover two objectives and key results [2]. Objectives describe what desired position and situation. The results key explains how this can be achieved. There are several characteristics of good key results, such as the quality that must be met by performance measurement indicators. It includes specific and time-bound, aggressive yet realistic, measurable, and verifiable [2]. This KRs are then shown in a

monitoring dashboard to determine what actions need to be done and insights that can be obtained to achieve these objectives [18].

However, not many research articles discuss this, especially in terms of use in the campus environment. Though, there is a common confusion. It is how different OKRs from KPIs. According to Radonic in [18], there are three distinguishing characteristics, namely: more frequent assessment, more transparent to all stakeholders, and goals establishment follows the bottom-up concept. Every stakeholder and person in charge has clear responsibilities and knows the overall KR within the organization's scope. Therefore, this study tries to adopt the OKR concept within the scope of PJJ implementation in higher education, which requires a set of guidelines on how to seamlessly adapt to this unfortunate and unpredicted situation and keep moving forward. Meanwhile, the campus can maintain the learning process's continuity, and the expected learning outcomes can be delivered.

III. METHODOLOGY

In this research, five steps are taken to build a simple Google Sheet-based dashboard by adopting the OKR concept. The first step is to identify the scope of the research and monitoring objectives using the dashboard. The research scope is learning activities, starting with filling in a study plan, providing materials, implementing exams, and assessments. It is also necessary to clarify the implementation characteristics, namely distance learning (PJJ) in pandemic conditions, and students who still have limited internet access.

The second step is to analyze the objectives and key results within this scope. The analysis was carried out based on evaluating the learning activities in the previous semester and concerning issues. The discussion involves every person in charge related with PJJ implementation in the institution. The objectives determination produces points of concern, used as a reference of how to conduct PJJ courses during the pandemic, what must be fulfilled and pursued regardless of the limited conditions. Each objective is then detailed into KRs, which transparently explains what needs to be achieved to fulfill the existing objectives. Each KR is equipped with a mapping of the data input type, which becomes the measurement variable presented in the dashboard.

The third step is to identify the data input mechanism, data structure, and data sources that serve as input media for measuring each KR's achievement. Each KR statement is also re-evaluated whether there are media that can supply data or not. After that, the various types of required data are grouped into multiple input types, where data collection can be done simultaneously, weekly, quarterly, and semester. The input types are also grouped based on how the data is obtained and who can input it. There are two strategies to gather data.

The first one uses Google Forms, allowing processing, and presenting real-time data on the monitoring dashboard. Then, it continues with importing CSV data, generated from the academic information system (AIS) and is carried out periodically. The data will be processed using the formula, prepared on the Google Sheet dashboard. Imported data is not presented in real-time but periodically according to the type of data. The import process uses a Python-based mini- program that retrieves the required data from AIS. The consideration is that AIS is still in the development phase, already has its workload, and runs. The dashboard solution can't interfere

with AIS's running, but still, be able to obtain the necessary data.

After defining OKR and data input types, the next step is to design a monitoring dashboard on the Google Sheet. At first, it starts with design on how each KR should be displayed, how to present concise and informative data, so that it would help the decision making to determine action and take insight. The design result is then followed by preparing the formula for calculating various input data via Google Form and importing CSV data using Python Script from related academic information system. The system architecture containing tools, designed modules, and actors is shown in Fig. 1.

Tools	Python Script, Academic IS	Google Form	Google Sheet		-
Modules	Fact finding		Data management	Visual user interface	Problem solving
Actors	System, operator	Online Assistant, Students	System, operator	System	Head of program

Fig. 1. System Architecture

Evaluation of the dashboard design and information visualization uses a cross-checking mechanism. The primary references as guidelines to conduct learning activities in a semester are PDDikti (national higher education database) reporting, study programs accreditation forms related to learning activities, as well as the results of calculations using last semester data. Thus, the monitoring dashboard design results can provide correct, practical information and still meet the existing standards.

IV. OKR ANALYSIS

The implementation of PJJ on each campus has different policies according to the student's characteristics. At the STT Nurul Fikri campus, some students still experience limited access to both internet quota and signal. Therefore, the primary guide for implementing distance learning is to provide learning methods and materials that students can access flexibly. Likewise, the attendance records and assignments also flexible, with the mechanism determined by each lecturer suiting their way in teaching. Lecturing activities consist of providing complete 14-course materials at the beginning of the semester. The purpose is to deliver the learning roadmap and conduct more discussion every week. In its implementation, these activities need a guide, supported by monitoring media so that actions to improve and resolve issues can be carried out effectively. In this study, OKR and monitoring dashboards are tools to help evaluate and enhance the learning process continuity.

Based on the previous semester evaluation and analysis, four objectives were defined. The first one ensures that all students follow all of the learning activities covering 14 topics and two assessments. The second is to deliver learning outcomes as defined and agreed upon regarding a pandemic situation. It then followed 100% compliance with national standards and reporting for the learning process. The last objective is to encourage students to build publishable creations regardless of the unfortunate and changing environment. The detailed key results are shown in the following tables.

Objective 1 is a set of goals to ensure the progress of learning and student participation. It is something to be

achieved in any condition so that students can still carry out their rights and obligations in learning. Rather than measuring motivation, which needs a more in-depth study, defining KR is carried out on observable actions as an impact of motivation to continue learning. Table 1 presents the KR, including student participation in the 14 learning topics and exams, discussions, and receiving learning materials.

TABLE I. KEY RESULT FOR ENSURING LEARNING CONTINUITY

<i>Code</i>	<i>Key Results</i>	<i>Input</i>
KR1	100% of each student attendance following 14 weeks	Type-6
KR2	100% participations in UTS & UAS	Type-5
KR3	0 case of three-times-in-a-row of absence	Type-6
KR4	active discussion forum within any choosen media high students participations (index ≥ 3 from 5)	Type-2
KR5	100% downloads number of the learning slides	Type-2
KR6	100% views number of the learning slides	Type-2

Objective 2 covers a set of goals regarding the quality of competencies expected in the implementation of learning. It is stated in the KR which represents the completeness of learning material, starting from the clarity of learning outcomes, the availability of material before lectures begin, to the level of understanding as measured by the achievement of student grades and satisfaction. It is shown in Table II.

TABLE II. KEY RESULT FOR DELIVERING LEARNING OUTCOME

<i>Code</i>	<i>Key Results</i>	<i>Input</i>
KR1	There is a definition of CPL, course topics (RPS), weight-contract since the beginning of the lecture	Type-2
KR2	There are slides and video material before the lecture takes place	Type-2
KR3	There are written reflection of each learning topics by all a students	Type-1
KR4	All students get score ≥ 75 of each exams	Type-2, 8
KR5	The results of the good assignment assessment, each students gets above ≥ 75	Type-2, 8
KR6	The results of the lecture understanding questionnaire	Type-1
KR7	Questionnaire results on lecturer attendance (discussion / feedback)	Type-3

Objective 3 covers the standards that must be met learning activities as a form of compliance with regulations. This objective also complements Objective 2, namely, the administration completeness and learning outcomes. The KR includes filling out study plans, filling in lecturers and students' presence, receiving material by students, and completing reporting. It is shown in Table III.

TABLE III. KEY RESULT FOR NATIONAL STANDARD COMPLIANCE

<i>Code</i>	<i>Key Results</i>	<i>Input</i>
KR1	The number of students with the contents of KRS $\geq$ active students in the previous seminars	Type-7
KR2	100% of students match the KRS filling	Type-7, 2
KR3	Completeness of student attendance	Type-6

KR4	Completeness of lecturer presence	Type-6
KR5	Complete and timely student assessments	Type-8
KR6	Questionnaire results on lecturer teaching satisfaction	Type-3
KR7	Students' average GPA ≥ 3.25	Type-8
KR8	The results of the campus support questionnaire for lectures is high (index ≥ 3 out of 5)	Type-3
KR9	3 9 Compatibility of the final scores at AIS and PDDikti	Type-8, 2

Objective 4 covers discussions related to providing soft skills for students to increase their adaptability to fast changes. It is manifested in the form of the results of the application of the various soft skills in the way of producing various creations, even in limited conditions. The KR in this objective includes students' ability to continue to collect assignments, participate in lecture discussions, and have work due to lectures and non-academic fields. It is shown in Table IV.

TABLE IV. KEY RESULT FOR ENCOURAGING STUDENTS CREATION

<i>Code</i>	<i>Key Results</i>	<i>Input</i>
KR1	100% of students continue to collect assignments and attend lectures if there are adjustments to learning tools	Type-8
KR2	$> 50\%$ of students can still attend discussions in various media	Type-2
KR3	Each course can recommend an outstanding output for further appreciation (a topic of competition / TA / research)	Type-4
KR4	10% of students produce work in the form of a published journal / paper / hki in 1 semester	Type-4
KR5	10% of students get appreciation from participating in competitions outside the campus in 1 semester	Type - 4

The measures of KR need various kinds of data input. Thus, eight input types consist of particular data columns in Google form and CSV. Input type 1 is for weekly evaluation data containing levels of learning material utilization, reflection, and lessons learned. Students fill in this input at the end of the week. The form is designed with only a few simple questions so that regularly filling it would not become burdensome for them. The purpose is to maintain the reliability of the results.

Input type 2 is an observation form filled in by online class assistants covering the class discussions, student attendance, completeness of learning material delivery, and acceptance. The first question asks how active participation in the class discussions, assessed in the scale 1 to 5 (1 is very quiet and 5 is lively discussion followed by hundreds of chats or more minutes in video conference discussion). The second and third ones count the numbers of downloads and views for each learning material and videos. They are both counted and updated per week, with expected numbers could reach the number of enrolled students. The fourth is about lecture delivery readiness covering defined learning outcomes and topics, adjusted to the pandemic conditions, and assessment contracts.

Input type 3 is a form filled in by students before mid-term and final term exams. This evaluation has been carried out before, only simplified, routinized, and processed. Previously, it was not clear how this data should be processed. The

questions gather information about results on lecturer attendance (discussion and feedback), teaching method satisfaction, and campus facility support satisfaction.

Input type 4 is a form asked about student observations, filled in by online lecture assistants from the results of observations and confirmation to related lecturers (academics), and taking data from associated units. It covers data gathering about outstanding student creations to be appreciated and followed up, like becoming the competition / Final Project/research topic, published in as conference/journal papers, to be claimed as an intellectual property right. The form also gathers information about competition appreciation for both academic and non-academic.

Input type 5 is an exam report form, filled in by the academic administration unit. It turns out that this data has the potential to be used in measuring student participation in important course milestones. If a student misses an exam, it is such an ample warning that it needs to be followed-up immediately. The form has data about the percentage of student participation in mid-test and end-of-semester test exams.

Input types 6 - 8 are the results of importing data from an academic information system (AIS), gathered using a mini-program. This program is built with Python and Jupiter to produce output in CSV form. It is then imported into calculation tables and formula templates prepared in Google Sheet. Data is generated at every defined milestone, namely beginning, week, midterm, and end of the semester. The information includes study plan entries, weekly attendance of lecturers and students, grades per half, and final semester.

V. DASHBOARD DESIGN

After defining the objectives and key results, calculating and structuring the data input, the next thing to do is designing the dashboard. The resulting dashboard is divided into three: weekly, semester, and detailed per student. The weekly dashboard, shown in Fig. 2, presents KR's measures for each course. There are two types of presentation, namely percentage and predicate label. They are accompanied by color formatting for easy action and insight determination to improve existing optimality. The percentage describes quantitative KR measures such as attendance, number of downloads, view slides, and completeness of weekly reflections. The predicate labels (O-Ok, W-warn, T-treat) are used for qualitative KR measures such as discussion activeness and level of student understanding based on written reflections. Green coloring indicates good, yellow indicates warnings to the related person in charge, and red requires immediate action to fix it.

Legend	Std	1	Weeks												
			A	D	S	V	R	U	2	3	5	6	7	UTS	UAS
A attendance	MK1	SI	200	95%	O	90%	30%	90%	O						
D discussion	TI	400	75%	W	80%	40%	80%	W							
S slides	SI	200	40%	T	30%	70%	75%	T							
V video	TI	400	40%	O	70%	40%	50%	O							
R reflection	MK3	SI	200	45%	O	50%	70%	5%	O						
U lev of understanding	TI	400	48%	W	40%	60%	30%	W							
Std students	MK4	SI	200	50%	T	90%	90%	30%	T						
O High & OK	TI	400	48%	O	30%	80%	40%	O							
W Warn	MK5	SI	200	100%	O	40%	50%	O							
T Treat	TI	400	48%	O	80%	70%	40%	O							
	MK6	SI	200	95%	W	100%	70%	70%	W						
	TI	400	48%	O	70%	60%	90%	O							
	MK7	SI	200	95%	W	60%	70%	30%	W						
	TI	400	75%	O	70%	60%	30%	O							

Fig. 2. Weekly Dashboard Example

The semester dashboard, shown in Fig. 3, presents KR measures where the calculation is conducted at the beginning, middle, and end of the semester. The presentation consists of checklists and predicate labels, also accompanied by color formatting. The checklists show course lecture readiness status, such as defined learning outcomes and 14 topics, agreement on assessment mechanisms, and acceptance of learning materials. The predicate label includes the student satisfaction questionnaire's result on the lecturers' presence in discussions and providing feedback, teaching methods for lecturers, and support for learning facilities.

		Status of Readiness in Beginning of Semester					Satisfaction Surveys				
		LO	Topics	Contract	Slide/Stat	Video/Stat	DA	DL	F	DA	DL
MK1	SI	☒	☒	☒	☒	☒	O	O	O	O	O
	TI	☒	☒	☒	☒	☒	W	O	T	W	O
MK2	SI	☒	☒	☒	☒	☒	T	O	W	T	O
	TI	☒	☒	☒	☒	☒	O	O	W	O	W
MK3	SI	☒	☒	☒	☒	☒	O	O	O	O	O
	TI	☒	☒	☒	☒	☒	W	O	O	W	O
MK4	SI	☒	☒	☒	☒	☒	T	O	W	T	O
	TI	☒	☒	☒	☒	☒	O	O	O	O	O
MK5	SI	☒	☒	☒	☒	☒	O	O	O	O	O
	TI	☒	☒	☒	☒	☒	O	O	O	O	O
MK6	SI	☒	☒	☒	☒	☒	O	W	W	O	W
	TI	☒	☒	☒	☒	☒	O	O	O	O	O
MK7	SI	☒	☒	☒	☒	☒	O	O	O	O	O
	TI	☒	☒	☒	☒	☒	O	O	O	O	O
Output to be recommend (end of semester)											
		Product	Paper	Academic Competition	Recommended						
MK1	SI	product A	international conf								
	TI		internal journal								
MK2	SI	product A		regional winner	another compet						
	TI										
High & OK											
Warn											
Treat											

Fig. 3. Semester Dashboard Example

The student dashboard, shown in Fig. 4, presents the learning outcomes per student and recapitulation. The dashboard shows information as a basis for action and treatment needed for certain cases with very poor grades. The form of presentation of information is in the form of a checklist and numbers. The checklist presents a summary of the report on the student's study plan conformity and the validity of the reported scores in national higher education databases. In numerical information, states the student assignments and exams scores and GPA.



Students Dashboard Distance Learning & Study Program

STT Terpadu Nurul Fikri

IPS avg per MK >= 3.2		3.1
IPK avg per MK >= 3.2		3.3
Total Mhs Aktif Sem lalu		380
Total Mhs Isi KRS		400
Avg Task Point		67

Students List														
NIM	Name	Stat KRS	Stat PDDikti	Avg Point for MK1			Avg Point for MK2			Avg Point for MK3				
				Task	UTS	UAS	Task	UTS	UAS	Task	UTS	UAS	IPS	IPK
1102001	Students A	☒	☒	75	80	70	75	80	70	75	80	70	3.5	3.4
1102002	Students B	☒	☒	80	90	80	80	90	80	80	90	80	3.8	3.76
1102003	Students C	☒	☒	89	50	89	89	50	89	89	50	89	3.2	3.16
1102004	Students D	☒	☒	60	80	80	60	80	80	60	80	80	2.75	2.9
1102005	Students E	☒	☒	70	80	70	70	80	70	80	70	70	3	3
1102006	Students F	☒	☒	80	30	80	80	30	80	80	30	80	3.7	3.6
High Warn Treat IPS average per MK >= 3.2														

Fig. 4. Students Dashboard Example

VI. DISCUSSION

One of the best practices of OKR compared to KPI is that it is derived based on the outcome. The objective statement should inspire anyone to know what's ahead and what they should do to achieve it. The four compiled objectives are to fulfill the four outcome dimensions that the study program must fulfill in carrying out education in the pandemic era. For each objective, a result key is defined as a reference for how the objective can be achieved. As objectives are defined, the key result must be clear in terms of form, how to measure it, and how to achieve it. By defining a clear key result, the

treatment and efforts to achieve it will become more evident. Several key result measures can be put together based on the timeline and form of presentation in the dashboard.

As for coloring, symbols, and representation of numbers using simple ones, namely checklists, percentages, number value points, GPA, and labels. The performance coloring is divided into three to make it easier to see and get insights for decision making. Thus, dashboard users can quickly monitor conditions and determine the treatment to adjust and improve the KR's achievement.

Several things need to be considered to implement this monitoring dashboard. There are new forms, and there must be operators to fill in the forms. This role can be assigned to the online learning assistants, whose task is to observe and fill in the observation forms. Also, re-use forms need coordination with related parties to adjust the data types and the filling timeline. The need for a third-party role is not to add burden to the lecturers and students, so they can focus on the learning process. The re-used forms are intended to reduce the new type of input so that the evaluation activity may run seamlessly, and the results can be processed. Many evaluations from the previous semesters have no exact format and template for data processing and presentation.

VII. CONCLUSION AND FUTURE RESEARCH

This research has produced a monitoring dashboard design with OKR using Google Sheet technology. This technology could visualize the required information related to measuring the KR achievement, with certain data input, calculated using spreadsheet formulas. Monitoring dashboard with Objective Key Result (OKR) may help optimize distance learning in the pandemic era by utilizing the definition of objectives and key results with adjustment and concerns regarding the situation. The learning activities monitoring is also carried out with the outcome as a guideline for what must be achieved. Thus, the learning process's implementation can remain focused and directed, even though various compromises regarding the uncertain situation.

Before it was applied to a real situation, there were two qualitative evaluations on the results, namely the OKR formulation and suitability of monitoring dashboard with campus policy makers' needs. The first was assisted by a practitioner who had successfully implemented OKR on a national scale organization. The second was assisted by a vice-dean who oversees the distance learning management. The test results are OKR is in accordance with the concept, as well as the dashboard design. There is a suggestion to integrate the design with learning management system data. This input can be used as a future research idea for the next development.

This dashboard design will be evaluated periodically by direct implementation in lecture activities in the following semester in 2020/2021. Thus, evaluating data input, formula validity, and business process mechanisms in providing continuous monitoring can be carried out according to a specific period. Continuous improvement is also applied to this dashboard with the main objective, presenting the valid information, up-to-date according to the period, and consistency of processes.

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